



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	ECO3121
Course title (English)	Introductory Econometrics
Course title (Chinese)	计量经济学导论
Units	3
Language of Instruction	English
Description (English)	This course provides an introduction to econometric theory and practice. We begin with a comprehensive discussion of simple and multiple regression analysis. More advanced topics will be tackled later in the semester, such as time series analysis, panel data, and binary models. The purpose of the course is to introduce you the statistical methods that economists use to test economic theory and interpret economic data, and to give you experience in estimating econometric models with real world data. This course has a strong applied emphasis. We will use STATA for data analysis. The course will provide you the opportunity to establish your skills of applying STATA in empirical study. Instructions on how to use the software will be given in tutorials and, if necessary, some of the tutorials may take place in the computer lab.
Description (Chinese)	本课程介绍了计量经济学的理论基础与实践。包括应用统计方法(主要通过回归分析)进行经济数据实证分析进而得出结论。本课的目的是为学生介绍计量经济学理论基础和相关实践,以及让学生学会如何使用真实数据进行计量经济模型分析。本课程具备很强的应用性,我们将使用 STATA 进行数据分析。本课程将培养学生在实证研究中运用 STATA 的技能。该软件的使用将在辅导课程中说明,一些习题课可能会安排在计算机实验室里进行。

2. Prerequisites / Co-requisites

A. Prerequisites

ECO2121 Basic Statistics or (STA2001 or STA2003)

B. Co-requisites



3. Learning Outcomes

By the end of this course, our student should be able to:

- 1) Critically read the quantitative analysis performed by other researchers.
- 2) Formulate their own econometric models to perform empirical investigations in business economics and finance
- 3) Collect and organize economic and financial data, and use statistical software packages (STATA) for empirical analysis

4. Course syllabus

N.A

5. Assessment Scheme

Component/ method	% weight
Assignments (4)	25
Midterm	30
Final Exam	45

6. Grade descriptor

General

Grade	Description
A	Outstanding performance on all learning outcomes
A-	Outstanding performance on most learning outcomes
B+,B,B-	Substantial performance on all learning outcomes, OR high
C+,C,C-	Substantial performance on the majority of learning outcomes
D+,D	Barely satisfactory performance on a number of learning outcomes
F	Unsatisfactory performance on a number of learning outcomes



7. Feedback for evaluation

(1) CTE, and (2) Informal feedback to instructor and/ or teaching assistant

8. Reading

Required Reading

1. Introductory Econometrics: A Modern Approach, 9781305270107, Jeffrey M. Wooldridge, Cengage Learning, 6, 2015, United States

9. Course components

Activity	Hours/week
Weekly lecture	3
Weekly tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Introduction of Econometrics and Statistics Review
002	Simple Linear Regression Model I
003	Multiple Linear Regression Model I: Estimation
004	Multiple Linear Regression Model II: Estimation
005	Multiple Linear Regression Model III: Inference
006	Multiple Linear Regression Model IV: Inference
007	Multiple Linear Regression Model IV: Further issues
008	Midterm
009	Instrument Variables
010	Binary Variables
011	Heteroskedasticity
012	Panel Data Analysis I
013	Panel Data Analysis II
014	Carrying Out an Empirical Project

11. Any other information



STATA:

Students are encouraged to use the statistical software STATA. Class examples will be illustrated using STATA. STATA is available at university computer lab, and the programming code for homework will be provided only for STATA users. However, if you want to use other programs, you are free to choose other statistical software for assignments.

No prior knowledge of STATA is assumed. Your TA will help you use this software package during the tutorial hours. However, a STATA introduction file can be found at <http://data.princeton.edu/stata/>, which is quite helpful for you to get a general idea of STATA.

Honesty in Academic Work:

The Chinese University of Hong Kong, Shenzhen places very high importance on honesty in academic work submitted by students, and adopts a policy of zero tolerance on cheating in examinations and plagiarism. Any related offence will lead to disciplinary action including termination of studies at the University. Please visit the following website for details of university policy on Honesty in Academic Work: <http://www.cuhk.edu.cn/departsite/ar/en/Academic.html>.

12. Version date

Version number	5
As of (date)	2024/06/21